CPSC 536Z Report

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1 Introduction

We study the Load-Balancing problem in the streaming model. An instance consists of a bipartite graph $G = (L, R, E)$ and the goal is to find the optimal assignment of $c : L \to R$ where $(l, c(l)) \in E$ (for $l \in L$) that minimizes $\max_{r \in R} |c^{-1}(r)|$. We also call the number of nodes in L assigned to $r \in R$, $c^{-1}(r)$, the load of r or $\textsc{load}_c(r)$. Thus, we want to find the assignment c that minimizes

$$\max_{r \in R}\{\textsc{load}_c(r)\}. \tag{1}$$

In the streaming model, the edges that arrive in a stream. And we want to find the optimal assignment with $\tilde{O}(n)$ space. The choice of $\tilde{O}(n)$ space is due to basic properties of G, such as connectivity, requiring $O(n)$ space to determine. We call an algorithm in the streaming model a *streaming algorithm.*

It is hard to compute the optimal solution of the Load-Balancing problem in the streaming model, so we usually hope to find α-approximations of the problem for $\alpha \geq 1$.

Definition 1.1 (α-approximation)**.** *For a minimization problem with optimal value* OPT*, an algorithm is an α-approximation for $\alpha \geq 1$ if it outputs a solution with value smaller than αOPT on any problem instance. A solution is also an α-approximation if it has value smaller than αOPT.*

For example, a streaming algorithm for the Load-Balancing problem is a 2-approximation if it always outputs an assignment where the max load at most 2OPT.

On graphs with perfect matchings (OPT $= 1$ by letting the assignment be a matching), a ϵ-approximation for $1 \leq \epsilon < 2$ requires that we output a perfect matching. Since an assignment that is not a perfect matching implies the existence of some node $r \in R$ with

$$\textsc{load}(r) \geq 2 > \epsilon = \epsilon \mathrm{OPT}. \tag{2}$$

Using lower bounds on perfect matchings in the streaming model, we know that $\Omega(\log(n))$ passes are required to compute a perfect matching [2]. Thus, we have a trivial lower bound of 2 on the approximation factor of any streaming algorithms for the Load-Balancing problem.

1.1 Connection to One-way Communication Complexity

The Load-Balancing problem has an analogous One-way Communication Complexity version. In this problem, Alice is given the bipartite graph $G_A = (L, R, E_A)$ and Bob $G_B = (L, R, E_B)$ such that $E_A \cap E_B = \emptyset$ and $E_A \cup E_B = E$. Namely, Alice and Bob partition the edges. Alice is allowed to communicate one message to Bob and Bob would output an assignment. We refer to this problem as the communication Load Balancing problem and let $\textsc{LoadBal}(n, \alpha)$ denote this problem for any G such that $|L| = n$ where the goal is to output an α-approximation. We also let $R^{\to}(\textsc{LoadBal}(n, \alpha))$ denote the randomized one-way communication cost of $\textsc{LoadBal}(n, \alpha)$ with success probability $\frac{2}{3}$.

The one-way communication model is intimately related to the streaming model. To see this, note that any streaming algorithm for a problem that uses k space defines a protocol of cost k: Alice can simply run the algorithm on her graph and sends the memory contents to Bob to complete the streaming algorithm. Therefore, if we can say that **"any (randomized) protocol for the communication Load-Balancing problem of cost $k = \tilde{O}(n)$ produces a $O(j)$-approximation"**, then no streaming algorithm that approximates Load Balancing to $\omega(j)$ can exist.

1.2 Coverage of this Report

We cover two main results in this area, each spanning one section. In Section 2, we study a class of graphs called α-*Matching Contractors*. It can be shown that when there is a 4α-Matching Contractor of size k, that $R^{\rightarrow}(\textsc{LoadBal}(n,\alpha)) = \Omega(\frac{k}{\log(n)})$ [1]. From there, the challenge of lower bounding α for protocols with cost at most $\tilde{O}(n)$ is reduced to the construction of suitable α-Matching Contractors.

In Section 3, we study a second class of graphs called α-*Sparsifiers*. These first appeared in literature to demonstrate upper bounds for the communication Load-Balancing problem, by deriving good protocols [3]. A more recent work demonstrates that α-Sparsifiers and α-Matching Contractors have a (almost) primal-dual relationship [1]. Combined with the result in Section 2, this cements α-Sparsifiers as the "correct" object to study the communication Load-Balancing problem.

It is worth nothing that to Professor Harms, the first result (Section 2) might be of greater interests since it is related to lower bounding problems, whereas the second result (Section 3) simply establishes a notion of equivalence between the complexity of the communication Load Balancing problem to the size of α-Sparsifiers with little implication on algorithms. The second result is of pure theoretical intrigue. Please contact me if something is unclear.

1.3 Notations

For some bipartite $G = (L, R, E)$ and $S \subseteq E$, we let $L(S) = \{l \in L : \text{for some } r \in R \text{ we have } (l, r) \in S\}$. We also let $N_G(X)$ denote the set of neighbours of X in the graph G. For $S \subseteq E$, we let $G \backslash S$ denote the graph $G = (L, R, E - S)$. Finally, $X \subseteq L, Y \subseteq R$, let $E(X, Y)$ denote the edges between X and Y in E.

2 Lower Bounding Streaming Complexity

2.1 Matching Contractors

We first define α-Matching Contractors.

Definition 2.1 (α-Matching Contractors)**.** *For $\alpha \geq 1$, a bipartite $G = (L, R, E)$ is an α-Matching Contractor iff E can be partitioned into matchings $M_1, ..., M_k$ such that $\forall i \in [k], L(M_i)$ has at most $\frac{|L(M_i)|}{\alpha}$ neighbours in $G \backslash M_i$.*

We let $\mathsf{MC}(n, \alpha)$ denote the maximum number of edges in a α-Matching Contractor $G = (L, R, E)$ with $|L| = n$. We also let $\mathsf{MC}(G, \alpha)$ denote the largest α-Matching Contractor H such that $H \subseteq G$.

2.2 Lower Bound on the Communication Load Balancing problem

Core to the result is the following theorem that relates $\mathsf{MC}(n, 4\alpha)$ to $\textsc{LoadBal}(n, \alpha)$.

Theorem 2.2 ([1])**.**

$$R^{\rightarrow}(\textsc{LoadBal}(n,\alpha)) = \Omega(\frac{1}{\log(n)}\mathsf{MC}(n, 4\alpha)) \tag{3}$$

To achieve this, we construct a hard input distribution.

2.2.1 Hard Input Distribution

Yao's original paper establishes the equivalence between the distributional and randomized one-way communication complexity [4]. Thus, it is sufficient to show lower bounds on a hard input-distribution for any deterministic one-way protocols to lower bound randomized one-way protocols.

Lemma 2.3 ([1])**.** *There exists a α-Matching Contractor with matchings $M_1, ..., M_k$ with $\Omega(\frac{1}{\log(n)}\mathsf{MC}(n, \alpha))$ edges such that for some constant r_0,*

$$\forall i \in [k], r_0 \leq M_i < \frac{4}{3} r_0. \tag{4}$$

Proof. Let G be a α-Matching Contractor with $\mathsf{MC}(n,\alpha))$ edges with matchings $\mathbf{M} = \{M_1, ..., M_k\}$. For $i \in \mathbf{N}$, let

$$G_i = \{M \in \mathbf{M} : \big(\frac{4}{3}\big)^i \leq |M| < \big(\frac{4}{3}\big)^{i+1}\}.$$

Note that only $O(\log(n))$ G_is are non-empty. So let $i_{max} = \arg\max_{i\in\mathbf{N}}\{|G_i|\}$. By a max over average argument, we have

$$|G_{i_{max}}| = \Omega(\frac{1}{\log(n)}\mathsf{MC}(n,\alpha)).$$

□

For the rest of this report, let $G_0 = (L_0, R_0, E_0)$ with $m_0 = |E_0|$ be such a graph from Lemma 2.3. Also let $M_1^0, ..., M_k^0$ from Equation (4). We define an *encoding graph* next, which encodes an m_0 bit string.

Definition 2.4 (Encoding Graph of G_0)**.** *Given $x \in \{0,1\}^{m_0}$, the encoding graph $G_x = (L, R, E_x)$ is the graph such that:*

1. *$L = L_0$*
2. *$R = R_0 \times \{0,1\}$. For $v \in R_0$, we denote the vertices in R_x as v_0 and v_1.*
3. *For every $e = (u,v) \in E_0$, $(u, v_0) \in E_x$ if $x_e = 0$. Otherwise $(u, v_1) \in E_x$ (if $x_e = 1$).*

For the rest of this report, let $G_x = (L, R, E_x)$ be the encoding graph of string x. It's important to see that each G_x is still a Matching-Contractors.

Lemma 2.5 ([1])**.** *$\forall x \in \{0,1\}^{m_0}, G_x = (L_x, R_x, E_x)$ is a 2α-Matching Contractor.*

Proof. For $i \in [k]$, construct M_i^x from M_i^0 by mapping each edge $e = (u,v) \in M_i^0$ to (u, v_0) if $x_e = 0$, otherwise (u, v_1). It's clear that $M_1^x, ..., M_k^x$ are matchings that partition E_x. Also note $|M_i^x| = |M_i^0|$.

Since G_0 is a 4α-contractor, we have by definition that

$$|N_{G_0\setminus M_i^0}(L(M_i^0))| \leq \frac{1}{4}|M_i^0|. \tag{5}$$

By construction of M_i^x, $N_{G_x\setminus M_i^x}(L(M_i^x)) \subseteq N_{G_0\setminus M_i^0}(L(M_i^0)) \times \{0,1\}$. So

$$|N_{G_x\setminus M_i^x}(L(M_i^x))| \leq |N_{G_0\setminus M_i^0}(L(M_i^0)) \times \{0,1\}| \leq \frac{1}{2}|M_i^0| = \frac{1}{2}|M_i^x|. \tag{6}$$

□

Note in particular that by construction, $L(M_i^x) = L(M_i^0)$. With these tools we can define the hard input distribution for the Communication Load Balancing problem.

Input Distribution μ: Let S be a set with $|L|$ nodes and P a perfect matching between L and S. Alice samples an $x \in \{0,1\}^{m_0}$ uniformly and receives the graph $(L, R \cup S, E_x)$. Bob samples $i \in [k]$ uniformly and receives the perfect matching $(L, R \cup S, P[L\setminus L(M_i^0)])$.

Note that Alice and Bob's inputs are independently and uniquely identified by x and i respectively. Note also that the perfect matching P is determined beforehand.

2.2.2 Proof of Lowerbound

The following observation is central to the breakthrough. For $x \in \{0,1\}^{m_0}$, let x_i denote the subsequence of bits for edges in M_i^x.

Lemma 2.6 ([1])**.** *Let π be a one-way deterministic protocol on input μ that is an α-approximation. On input $(x,i) \sim \mu$ for which π outputs the correct answer, at least half the values of x_i are deterministically fixed given only $\pi(x)$ and i.*

Proof. Consider $(x, i) \sim \mu$. Note that by matching edges of $L \backslash L(M_i^x)$ according to P and edges of M_i^x according to G^x, that we have an assignment with load 1 (it is a matching). So any α-approximate protocol π produces an assignment such that $\max \text{LOAD}_{r \in R_x \cup S}(r) \leq \alpha \text{OPT} = \alpha$.

Consider the nodes $S \subseteq L(M_i^x)$ that don't use the edges in M_i^x. Since G_x is an 2α-matching contractor, S is assigned to $\leq \frac{1}{2\alpha}|M_i^x|$ neighbours. By an average over max argument, the load of same node is at least

$$\frac{|S|}{\frac{1}{2\alpha}|M_i^x|} \leq \alpha \implies |S| \leq \frac{1}{2}|M_i^x|. \tag{7}$$

This means that whenever Bob's output is correct, it contains at $\frac{1}{2}|M_i^x|$ from M_i^x. Each edge uniquely identifies the corresponding bit in x. So whenever the output is correct, Bob needs to determine at least half the values of x_i. $\square$

This Lemma tells us that if the output is correct, then Bob must know a good amount of information about Alice's input. However, since Alice knows nothing about the Bob's input, she must communicate "a lot" to give Bob enough information. We formalize this next.

We let X, I, Π denote the random variable of Alice's input, Bob's input, and the protocol's message dependent on Alice's input. Also let X_I be the subsequence of X that corresponds to the edges in M_I^x.

Lemma 2.7 ([1]).

$$H(X_I \mid \Pi, I) \leq \frac{8}{9}r_0 + 1$$

Proof. Let Z be the indicator variable of whether Bob's output is an α approximation or not.

$$\begin{aligned}
& H(X_I \mid \Pi, I) && && (8)\\
&= H(X_I, Z \mid \Pi, I) - H(Z \mid \Pi, I, X_I) && \to \text{(chain rule)} && (9)\\
&= H(X_I \mid \Pi, I, Z) + H(Z \mid \Pi, I) - H(Z \mid \Pi, I, X_I) && \to \text{(chain rule)} && (10)\\
&\leq H(X_I \mid \Pi, I, Z) + 1 && \to \text{(Lemma A.1)} && (11)\\
&= P(Z=1)H(X_I \mid \Pi, I, Z=1) + P(Z=0)H(X_I \mid \Pi, I, Z=0) + 1 && && (12)\\
&\leq P(Z=1)\frac{1}{2}(\frac{4}{3}r_0) + P(Z=0)(\frac{4}{3}r_0) + 1 && && (13)\\
&\leq \frac{2}{3}\frac{1}{2}(\frac{4}{3}r_0) + \frac{1}{3}(\frac{4}{3}r_0) + 1 && && (14)\\
&= \frac{8}{9}r_0 + 1 && && (15)
\end{aligned}$$

With (13) due to Lemma 2.6 with the fact that X_I is a substring of length at most $\frac{4}{3}r_0$. $\square$

Lemma 2.8 ([1]).

$$H(X_I \mid \Pi, I) \geq r_0 - \frac{1}{k}||\pi||$$

where $||\pi||$ is the cost of the protocol.

Proof.

$$
\begin{aligned}
& H(X_I \mid \Pi, I) && (16)\\
&= \sum_{i=1}^{k} P(I=i) H(X_i \mid \Pi, I=i) && (17)\\
&= \frac{1}{k}\sum_{i=1}^{k} H(X_i \mid \Pi, I=i) \quad \rightarrow \text{(uniform distribution of } I) && (18)\\
&= \frac{1}{k}\sum_{i=1}^{k} H(X_i \mid \Pi, I) \quad \rightarrow \text{(independence of } X_i \text{ and } \Pi(X) \text{ to } I) && (19)\\
&\geq \frac{1}{k}\sum_{i=1}^{k} H(X_i \mid \Pi, I, X_1, ..., X_{i-1}) \quad \rightarrow \text{(conditioning decreases entropy)} && (20)\\
&= \frac{1}{k}(H(X \mid \Pi)) && (21)\\
&\geq \frac{1}{k}(H(X) - H(\Pi)) \quad \rightarrow \text{(chain rule + non-negativity)} && (22)\\
&\geq \frac{1}{k}(m_0 - ||\pi||)(X \text{ uniform}) && (23)
\end{aligned}
$$

Finally, note that $m_0 \geq kr_0$ by construction. So

$$H(X_I \mid \Pi, I) \geq \frac{1}{k}(m_0 - ||\pi||) \geq r_0 - \frac{1}{k}||\pi|| \tag{24}$$

as required. □

We combine the two bounds to obtain Theorem 2.2.

Proof (of Theorem 2.2). For any protocol π, we use Lemma 2.7 and 2.8 to obtain

$$||\pi|| \geq \frac{1}{9}kr_0 - k. \tag{25}$$

The case where $\mathsf{MC}(n, 4\alpha) = O(n \log n)$ is trivial. In this case, give Alice a random a perfect matching and Bob the empty graph. Then Alice must communicate $\Omega(n \log n)$ bits to send one edge per vertex to Bob for any finite approximation (adapting the convention that invalid assignments have infinite load). So Equation (3) holds. When $\mathsf{MC}(n, 4\alpha) = \omega(n \log n)$, we have that $r_0 = \omega(1)$. Then $kr_0 = \omega(k)$ so

$$||\pi|| \geq \frac{1}{9}kr_0 - k \approx \frac{1}{9}kr_0 = \Omega(m_0) = \Omega(\frac{1}{\log(n)}\mathsf{MC}(n, \alpha))$$

as required. □

2.3 Lower-bounding the Streaming Model

The final piece of the puzzle is the construction of *large* Matching-Contractors. Assadi et al. show the following

Theorem 2.9 (Dense Matching-Contractors[1]). *For $\epsilon \in (0,1)$ sufficiently small, and $n \geq 1$ sufficiently large, there are ($n^{\frac{1}{4}-O(\epsilon)}$)-Matching Contractors with $n^{1+\Omega(\epsilon^2)}$ edges.*

By, setting $\alpha = n^{\frac{1}{4}-O(\epsilon)}$, Theorem 2.2 and 2.9 tell us

$$R^{\rightarrow}(\textsc{LoadBal}(n, n^{\frac{1}{4}-O(\epsilon)})) = \Omega(\frac{1}{\log(n)}\mathsf{MC}(n, 4n^{\frac{1}{4}-O(\epsilon)})) \geq \Omega(\frac{1}{\log n}n^{1+\Omega(\epsilon^2)}).$$

Note that in the last term, the $\log n$ factor is subsumed by the constant in the power.

$$\Omega(\frac{1}{\log n} n^{1+\Omega(\epsilon^2)}) = \Omega(\frac{1}{\log n} n^{1+c}) = \Omega(\frac{n^{\frac{c}{2}}}{\log n} n^{1+\frac{c}{2}}) = \Omega(n^{1+\frac{c}{2}}) = \tilde{\omega}(n).$$

So we would need to send more than $\tilde{O}(n)$ bits. This gives us the follow Corollary on the streaming model.

Corollary 2.9.1 ([1])**.** *There is no streaming algorithm for obtaining a $n^{\frac{1}{4}-o(1)}$ approximation to the load-balancing problem with probability of success at least $\frac{2}{3}$.*

3 Equivalence of Sparsifiers to Load-Balancing

3.1 Sparsifiers and Upper Bounds

Definition 3.1 (α-Sparsifier)**.** *Let $G = (L, R, E)$ be a bipartite graph and $\alpha \geq 1$. $H = (L, R, E_H) \subseteq G$ is a α-Sparsifier iff for every set $C \subseteq L$,*

$$\text{OPT}\textsc{load}(H[C \cup R]) \leq \alpha \cdot \text{OPT}\textsc{load}(G[C \cup R])$$

We let $\textsc{sparsifier}(G, \alpha)$ denote the minimum possible number of edges in a α-Sparsifier of G, ad $\textsc{sparsifier}(n, \alpha)$ is the maximum value of $\textsc{sparsifier}(G, \alpha)$ over all $G = (L, R, E)$ with $|L| = n$. α-Sparsifiers can be used to give upper bounds on $\tilde{R}(\textsc{LoadBal}(n, \alpha))$. Sparsifiers have another characterization that are easier to work with.

Definition 3.2 (α-Sparsifiers (2)[1])**.** *Let H be an α-Sparsifer of G. Then for every $X \subseteq L$ such that there is a matching in G containing X*

$$|N_H(X)| \geq \frac{1}{\alpha}|X|.$$

We call such an X matchable.

Theorem 3.3 ([[1])**.** *] Let $n, \alpha \geq 1$. Suppose $\textsc{sparsifier}(n, \alpha) = T$. Then there is a deterministic one-way protocol π for $\textsc{LoadBal}(n, \alpha + 1)$ with com cost $||\pi|| = O(T \log(n))$ bits.*

Proof.

Protocol. Alice computes α-Sparsifier H_A of her input G_A and sends it to bob. Each edge requires $\log(n)$ bits to send. Bob outputs the optimal assignment on $G' = H_A \cup G_B$.

We want to show that

$$\text{OPT}\textsc{load}(G') \leq (\alpha + 1)\text{OPT}\textsc{load}(G)$$

Let A^* be an optimal assignment in G, with edges E^*. Let $E_A^* := G_A \cap E^*$ and $E_B^* := G_B \cap E^*$. E_A^* defines an assignment on $L_A := L(E_A^*)$, which we denote A_A^*. Then

$$\text{OPT}\textsc{load}(G_A[L_A \cup R]) \leq \textsc{load}(A_A^*) \leq \textsc{load}(A^*). \tag{26}$$

So by definition of α-Sparsifier, there is an assignment A'_A of $H_A[L_A \cup R]$ with

$$\textsc{load}(A'_A) \leq \alpha\textsc{load}(A_A^*) \leq \alpha\textsc{load}(A^*)$$

Finally, we construct an assigment A'. For $x \in L_A$, map $A'(x) = A'_A(x)$. Otherwise, map $A'(x) = A_B^*(x)$ (A_B^* defined symmetrically to A_A^*). All edges of A' are contained in G' and

$$\textsc{load}(A') \leq \textsc{load}(A'_A) + \textsc{load}(A_B^*) \leq (\alpha + 1)\textsc{load}(A^*). \tag{27}$$

□

Thus, similar to how finding a lower bound on $\tilde{R}(\textsc{LoadBal}(n,\alpha))$ is reduced to finding α-Matching Contractors, finding an upper bound is reduced to finding small α-Sparsifers. Konrad et al. are able to show that $\textsc{sparsifier}(n, n^{\frac{1}{3}}) \leq 2n$. Namely, that every graph $G = (L, R, E)$ with $|L| = n$ admits a $n^{\frac{1}{3}}$-Sparsifier with size at most $2n$. Thus, we are able to achieve a $(n^{\frac{1}{3}} + 1)$-approximation by communicating using $O(n \log(n))$ bits to communicate the sparsifier. Assadi et al. show that the equivalence goes in the other direction.

Theorem 3.4 ([1]). *Suppose there exists a (randomized) communication protocol π for* $\textsc{LoadBal}(n,\alpha)$ *with cost C and probability of success $\frac{2}{3}$, then*

$$\textsc{sparsifier}(n, 8\alpha) = O(C \log^2(n)). \tag{28}$$

This theorem is shown by establishing a near primal-dual relationship between α-Sparsifers and α-Matching Contractors. Theorem 2.2 then completes the equivalence.

3.2 Sparsifiers and Matching Contractors

Lemma 3.5 ([1]).

$$\mathsf{MC}(n, 2\alpha) = O(\textsc{sparsifier}(n, \alpha)) \tag{29}$$

Proof. Let G be a 2α-Matching Contractor with $\mathsf{MC}(n, 2\alpha)$ edges, and let H a α-Sparsifier of G. Consider any matching M_i of G, with left endpoints L_i and right R_i. Suppose $|E(H) \cap M_i| < \frac{1}{2}|M_i|$. Let L_i' be the points in L whose matching edge in M_i is in not in H. Then $|L_i'| > \frac{|L_i|}{2}$ by our assumption. Using definition of Matching Contractors, it follows that

$$|N_H(L_i')| \leq |N_{G \setminus M_i}(L_i)| \leq \frac{1}{2\alpha}|L_i| < \frac{1}{\alpha}|L_i'|, \tag{30}$$

a contradiction to Definition 3.2 since L_i' is matchable (by M_i). □

Next, we demonstrate the primal dual relationship. We define a linear program, denoted $\mathsf{LP}(G, \alpha)$, for bipartite G and $\alpha \geq 1$. For $X \subseteq L, Y \subseteq R$, we (X, Y) is a contracting pair if X is matchable and $|Y| \leq \frac{1}{2\alpha}|X|$.

$$\min \sum_{e \in E} x_e \tag{31}$$

$$\sum_{e \in E(X, R \setminus Y)} x_e \geq \frac{|X|}{2} \text{ for all contracting pairs } (X, Y) \tag{32}$$

$$x_e \geq 0, e \in E \tag{33}$$

Lemma 3.6 ([1]).

$$\mathsf{LP}(G, \alpha) \leq \textsc{sparsifier}(G, \alpha) \tag{34}$$

Proof. Let H be a α-Sparsifier of G with $\textsc{sparsifier}(G, \alpha)$ edges. We show x where $x_e = 1$ for $e \in E(H)$ and 0 otherwise is a feasible solution to $\mathsf{LP}(G, \alpha)$. Consider any contracting pair (X, Y). Define

$$X' := \{v \in X | N_H(v) \subseteq Y\}.$$

In particular, $N_H(X') \subseteq Y$. Note that if $|X'| > \frac{1}{2}|X|$, we have

$$|X'| > \frac{1}{2}|X| \geq \alpha|Y| \Rightarrow \frac{1}{\alpha}|X'| > Y. \tag{35}$$

This contradicts Definition 3.2 for H, since X' is matchable (in turn because X is matchable). Thus, at least $\frac{|X|}{2}$ vertices of X have a neighbour in $R \setminus Y$. Thus, (32) is satisfied. □

Lemma 3.7 ([1]).

$$\textsc{sparsifier}(G, \alpha) \leq 20 \ln(n) \mathsf{LP}(G, \frac{\alpha}{2}) \tag{36}$$

For $G = (L, R, E)$ with $|L| = n$, $|R| \leq n^2$, and $\alpha \geq 2$.

Proof. Consider a feasible x to $\mathsf{LP}(G, \frac{\alpha}{2})$. Define $p = 10\ln(n)$. We perform randomized rounding. For every $e \in G$, add e to H with probability $\min\{px_e, 1\}$. Let E_H be the edges of H.

$$\mathbb{E}[|E_H|] = \sum_{e\in E} \min\{px_e, 1\} \leq \sum_{e\in E} px_e = \mathsf{LP}(G, \frac{\alpha}{2}). \tag{37}$$

By Markov Inequality,

$$P[|E_H| \leq 20\ln(n)\mathsf{LP}(G, \frac{\alpha}{2})] \geq \frac{1}{2}. \tag{38}$$

We want to argue WHP, H satisfies Definition 3.2. Equivalently, WHP, for any matchable X and $Y \subseteq R$ with $|Y| < \frac{1}{\alpha}|X|$, we have $E_H(X, R\backslash Y) \neq \emptyset$. This implies that $N_H(X) \neq Y$. For any such a pair of X and Y, let $Z_{X,Y}$ be the event that $E_H(X, R\backslash Y) = \emptyset$.

Consider first a pair X, Y such that $|X| = k$. Note we are considering $\mathsf{LP}(G, \frac{\alpha}{2})$, so (X, Y) forms a contracting pair. By feasibility of x, (32) gives us that

$$\sum_{e\in E(X,R\backslash Y)} x_e \geq \frac{k}{2}. \tag{39}$$

If any $e \in E(X, R\backslash Y)$ has $px_e \geq 1$, then $e \in H$ and we are done ($E_H(X, R\backslash Y) \neq \emptyset$ as required). So assume $px_e < 1$ for all $e \in E(X, R\backslash Y)$. Since each e is sampled independently,

$$P[Z_{X,Y}] \tag{40}$$

$$= \prod_{e\in E(X,R\backslash Y)} (1 - px_e) \tag{41}$$

$$\leq \exp(-\sum_{e\in E(X,R\backslash Y)} px_e) \tag{42}$$

$$\leq \exp(-\frac{pk}{2}) = n^{-5k} \tag{43}$$

with the last inequality due to (39). For $k \in \{1, ..., n\}$, the number of such pairs (X, Y) is at most $n^k|R|^k \leq n^{3k}$. Then,

$$P[\bigcup_{k\in[n]} \{ \bigcup_{X,Y \text{contracting},\ |X|=k} Z_{X,Y}\}] \leq n \cdot n^{3k} \cdot n^{-5k} \leq n^{1-2k} \leq \frac{1}{n}$$

So with probability at least $1 - \frac{1}{n}$, H satisfies Definition 3.2. Hence by the probabilistic method, there is an α-Sparsifier of the desired size. $\square$

Using a similar randomized rounding approach, Assadi et al. show

Lemma 3.8 ([1])**.**

$$\mathsf{LP}(G, \alpha) \leq 20\mathsf{MC}(G, \alpha) \tag{44}$$

Overall, Lemma 3.5, 3.6, 3.7, 3.8 give us the following theorem.

Theorem 3.9 ([1])**.**

$$\mathsf{MC}(n, 2\alpha) = O(\textsc{sparsifier}(n, \alpha)) = O(\mathsf{MC}(n, \frac{\alpha}{2})\ln(n)) \tag{45}$$

In turn, this can be combined with Theorem 2.2 to obtain Theorem 3.4.

References

[1] Sepehr Assadi, Aaron Bernstein, Zachary Langley, Lap Chi Lau, and Robert Wang. Streaming and communication complexity of load-balancing via matching contractors, 2024.

[2] Venkatesan Guruswami and Krzysztof Onak. Superlinear lower bounds for multipass graph processing, 2016.

[3] Christian Konrad and Adi Rosén. Approximating semi-matchings in streaming and in two-party communication, 2013.

[4] Andrew Chi-Chin Yao. Probabilistic computations: Toward a unified measure of complexity. In *Proceedings of the 18th Annual Symposium on Foundations of Computer Science*, SFCS '77, page 222–227, USA, 1977. IEEE Computer Society.

A Information Theory

Lemma A.1 (Entropy of Indicator Variable)**.** *For an indicator variable Z,*

$$0 \leq H(Z) \leq 1$$

Proof. The lower bound is by non-negativity of entropy. We also know that entropy is maximized when the distribution is uniform. So

$$H(Z) = p \log_2(\frac{1}{p}) + (1-p) \log_2(\frac{1}{1-p}) \leq 1$$

by letting $p = \frac{1}{2}$. □